

Meghana M
Sr. Python Developer
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SUMMARY:

- Full Stack Python Developer with 9+ years of experience delivering scalable applications in **Python (Django, Flask, FastAPI), ReactJS**, and **SQL (PostgreSQL/MySQL)**. Skilled in **RESTful APIs, OAuth2.0/JWT security, and CI/CD pipelines** with cloud deployment on **AWS/Azure**. Strong expertise in **FastAPI, AWS serverless (Lambda, DynamoDB, S3), Terraform (IaC), Docker, GitHub, and CI/CD pipelines**, delivering scalable backend services and platform engineering solutions.
- Experienced in working with various stages of Software Development Life Cycle (SDLC), Software Testing Life Cycle (STLC) and QA methodologies from project definition to post-deployment documentation.
- Experience with Design, code, debugging operations, reporting, data analysis and Web Applications utilizing Python.
- Customized pytest plugins and hooks to enhance test execution and reporting.
- Developed custom reports using HTML, Redshift, Streamlit Analytics AWS, Python and MySQL
- Experience in using pip command to install, manage, and utilize different python libraries like Beautiful Soup, NumPy, Pillow, Django rest framework, Django rest framework-jwt, pyspark, python-twitter, Celery, Matplotlib, and used PEP8 coding convention.
- Developed and secured OAuth2 providers and clients using frameworks like FastAPI, Django, or Flask.
- Good experience in developing web applications implementing MVT/MVC architecture using Django, Flask, Webapp2 and spring web application frameworks, with good understanding of Django ORM and SQL Alchemy
- Responsible for configuring a Python REST API framework utilizing Flask and FastAPI, as well as delivering interactive Open API Standard (aka. Swagger) API documentation to other cross-functional teams.
- Experienced in PHP, MySQL, OOPS, Design Patterns, HTML5, CSS, C#, Python and Django
- Knowledgeable in Node.js, and React.js, Responsive web design.
- In-Depth understanding of Spark Architecture including Spark Core, Spark SQL, Data Frames, Spark Streaming with PySpark and PANDA library.
- Experience in big data processing pipelines using AWS EMR with Python, PySpark, and Hadoop.
- Experienced in implementing Object Oriented Python, Hash Tables (Dictionaries) and Multithreading, Django, MYSQL, Exception Handling and Collections using Python.
- Extensive experience in developing web applications using Python, Django, Flask, Pyramid, Sanic frameworks. worked with several python libraries including Beautiful Soup, Twisted, NumPy, matplotlib, SciPy, PyQt, Scapy, SQL Alchemy, Boto, logging, Nose, Blaze, Hug.
- Strong experience of software development in Python (libraries used: libraries - Beautiful Soup, NumPy, SciPy, matplotlib, python-twitter, Panda's data frame, networks, urllib2, MySQL dB for database connectivity) and IDEs - sublime text, Spyder, emacs.
- In-Depth understanding of Spark Architecture including Spark Core, Spark SQL, Data Frames, Spark Streaming with PySpark and LLMs PANDA library.
- Experience in using Design Patterns such as MVC, Singleton and frameworks such as DJANGO, Ability in handling Django ORM (Object-Relational Mapper) and SQL Alchemy.
- Expertise in developing Restful micro services using python and fast Api frameworks
- Experience in using Version Control Systems like GIT, SVN and CVS to keep the versions and configurations of the code organized.
- Integrated factory boy with Django ORM / SQL Alchemy to streamline the creation of test objects in large-scale applications.
- Designed and implemented ETL workflows on EMR to process large-scale datasets efficiently
- Exposure to CI/CD tools - Jenkins for Continuous Integration, Ansible for continuous deployment.
- Experience in maintaining and executing build scripts to automate development and production builds.
- Experience in Amazon Web Services (AWS) cloud platform like EC2, Virtual private clouds (VPCs), Storage models (EBS, S3, instance storage), Elastic LoadBalancers (ELBs).

- Designed the data models to be used in data intensive AWS Lambda applications which are aimed to do complex analysis creating analytical reports for end-to-end traceability, lineage, and definition of Key Business elements from Aurora.
- Implemented automated local user provisioning instances created in AWS cloud and google cloud.
- Excelled on creating AMI (AWS Machine Images) that utilizes ELB (Elastic Load Balancer) and Auto Scaling. Experience with designing and configuring secure Virtual Private Cloud (VPC) through private and public networks in AWS by creating various subnets, routing table, Network ACL, NAT gateways.
- Implemented a 'server less' architecture using API Gateway, Lambda, and Dynamo DB and deployed AWS.
- Code from Amazon S3 buckets. Created a Lambda Deployment function and configured it to receive events from your S3 bucket.
- Experience in deploying applications in heterogeneous Application Servers TOMCAT, Web Logic and Oracle Application Server.
- Used Pydantic for fast and type-safe data parsing and validation in FastAPI applications.
- Hands on experience in using NOSQL libraries like MongoDB, Cassandra, Redis and relational databases like Oracle, SQLite, PostgreSQL and MYSQL databases.
- Experience in deploying applications in heterogeneous Application Servers TOMCAT, Web Logic and Oracle Application. Server.

Education: Bachelor's in Information technology, GITAMS University, India. 2014

Technical Skills:

Python Development: Python, Django, Flask, Flask-SQLAlchemy, Pyramid, JavaScript, TypeScript, RESTful APIs, GraphQL, Microservices, Node.js, FastAPI, Kafka, Avro, SOAP, WSDL, Internal Tooling (profiling, linting, automated testing, observability).

Cloud: AWS, Azure, Google Cloud Platform (GCP).

AWS Services: Lambda, EC2, S3, RDS, DynamoDB, Cloud Formation, IAM, Cloud Watch, API Gateway, Glue, Redshift, ECS, EKS.

Data and Analytics: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Big Query, Tableau, Power BI, AWS Quick Sight

DevOps and Deployment: Docker, Kubernetes, OpenShift, Terraform, Ansible, Jenkins, Maven, ANT, AWS Code Pipeline, Azure DevOps, CloudFormation, GitHub Actions.

Databases: DB2, SQL, PL/SQL, MongoDB, MySQL, PostgreSQL, Cassandra

Version Control & Servers: Git, SVN, GitHub, Bitbucket, WebLogic, Nginx, Apache, JBoss.

Monitoring and Logging: AWS CloudWatch, Azure Monitor, Prometheus, Grafana, ELK

Security and Authentication: JWT Authentication, SSL/TLS, OAuth, OWASP, Azure AD

Agile and Project Management: Scrum, Kanban, JIRA, Confluence, Sprint Planning, Daily Stand-ups, Retrospectives

Testing and Quality Assurance: PyUnit, Jest, Mocha, PyTest, Mockito, Selenium

Other Tools and Technologies: IntelliJ IDEA, Eclipse IDE, PyCharm, VS Code, Kafka, RabbitMQ, Informatica, Linux.

WORK EXPERIENCE:

Client: Charles Schwab, Dallas, TX

June 2024 – Till Date

Sr. Python Developer

Responsibilities:

- Designed and developed **RESTful APIs** with FastAPI/Python, integrated with **PostgreSQL/MySQL**, and secured with **OAuth2.0/JWT authentication**

- Built responsive **ReactJS** components and integrated them with backend APIs to deliver seamless user experiences for financial applications.
- Implemented **CI/CD pipelines** with Jenkins and GitHub Actions, automated testing with **PyTest and Jest**, reducing release cycle time by 30%.
- Implemented efficient software development life cycle (SDLC) methodologies to ensure smooth project delivery and meet business objectives.
- Implemented **profiling, linting, and automated testing practices** to ensure clean, maintainable, and high-performance backend code.
- Integrated real-time data streaming and processing functionalities using Kafka, ensuring timely and accurate data handling.
- Utilized TypeScript and Node.js to develop scalable and maintainable server-side applications, enhancing code readability and developer productivity.
- Managed relational databases like MySQL and NoSQL databases like MongoDB for data storage, retrieval, and manipulation in diverse application scenarios.
- For intricate user input procedures, dynamic reactive forms with unique validators were designed.
- Designed and implemented cloud-based solutions on AWS (EC2, Lambda, RDS, S3, ELB, CloudFront) for scalable and reliable application hosting and data storage.
- Applied machine learning techniques using PyTorch for data analysis, modeling, and predictive analytics, driving actionable insights and business decisions.
- Created interactive and insightful visualizations using Tableau, enabling stakeholders to derive meaningful insights from data.
- Leveraged NumPy and Pandas for data manipulation, analysis, and transformation, handling large datasets efficiently.
- Automated infrastructure provisioning and management using Terraform, Nginx, and Docker, optimizing deployment processes and reducing manual errors.
- Configured continuous integration and continuous deployment (CI/CD) pipelines with Jenkins, ensuring rapid and reliable software delivery.
- Orchestrated containerized applications and microservices using Kubernetes, improving scalability, fault tolerance, and resource utilization.
- Monitored system performance and health using Prometheus and Grafana, implementing proactive measures for optimizing application performance.
- Collaborated with teams using Bitbucket for version control, code reviews, and streamlined development workflows.
- Conducted comprehensive unit testing with PyTest to ensure code quality, reliability, and maintainability.
- Implemented JSON Web Tokens (JWT) for secure authentication and authorization in web applications, ensuring data privacy and integrity.
- Practiced Agile and Scrum methodologies for iterative and incremental development, fostering collaboration and adaptability.
- Managed AWS RDS for relational database services, AWS S3 for scalable object storage, and AWS ELB for load balancing, ensuring high availability and performance.
- Implemented content delivery networks (CDNs) using AWS CloudFront for fast and reliable content delivery, improving user experience.
- Managed containerised workloads with Kubernetes using Anthos GKE, ensuring high availability and efficient resource utilisation in both on-prem and cloud environments.
- Automated infrastructure provisioning and application management across Google Cloud, AWS, and Azure using Anthos, enabling enhanced consistency and governance.
- For smooth application deployment and scalability across hybrid and multi-cloud infrastructures, Google Cloud's Anthos was used to implement and manage multi-cloud environments.
- Optimised cloud-native applications by leveraging Anthos Service Mesh for improved microservices communication, security, and observability across different cloud platforms.
- Configured and oversaw the Azure Container Registry (ACR) to safely store and administer Docker container images.
- For smooth container deployment, ACR was integrated with Azure DevOps CI/CD pipelines and Azure Kubernetes Service (AKS).
- ETL processes for handling massive datasets on Apache Spark clusters were put into place.

- AWS CloudWatch Logs, Alarms, and X-Ray for tracing Python microservices were used to monitor application logs and create alarms.
- used Python applications' SSM Parameter Store and AWS Secrets Manager to securely handle secrets.

Environment: Fast API, Python 3.9, MySQL, MongoDB, RESTful APIs, AWS, PyTorch, Terraform, Nginx, Maven, Eclipse, PyCharm, Informatica, Jenkins, Kubernetes, Docker, Grafana, Bitbucket, PyTest, JWT, NumPy, Pandas.

Client: Visa, Austin, TX

Oct 2022 – Jun 2024

Role: Sr. Python Developer

Responsibilities:

- Designed and optimized SQL/NoSQL queries in Java for efficient data handling in PostgreSQL and MongoDB.
- Integrated OAuth 2.0 and JWT, for secure user authentication and authorization across digital banking channels and internal applications.
- Enhanced backend services with **FastAPI and Flask**, implementing **role-based authentication with OAuth/JWT** for secure banking APIs.
- Integrated GraphQL with existing REST and SQL/NoSQL backends, enabling unified data access layers.
- Performed container health checks, auto-scaling, and rolling updates to maintain high service availability.
- Integrated FastAPI with SQLAlchemy and Pydantic for robust data validation and ORM support.
- Developed responsive and user-friendly frontend interfaces using React.js and Vue.js for digital onboarding, KYC verification, and loan application workflows.
- Deployed Airflow on Docker and Kubernetes environments to scale and manage Python-based data pipelines in production.
- Developed custom Python operators and sensors in Apache Airflow to support complex workflow logic across data engineering tasks.
- Configured AWS IoT rules to route messages to Lambda, S3, and DynamoDB for real-time processing and storage.
- Developed modern, responsive web interfaces using HTML5, CSS3, and JavaScript (ES6+), adhering to WCAG accessibility standards and optimizing for cross-browser compatibility and mobile-first design.
- Automated data quality checks using custom Python sensors and operators in Airflow, ensuring data consistency and reliability.
- Implemented containerized Python services using Docker and Kubernetes (EKS) to support dynamic scaling for peak banking operations and disaster recovery.
- Created batch data processing jobs using Pyspark and Python for transaction categorization, fraud detection models, and customer segmentation.
- Leveraged AWS S3, Lambda, RDS (PostgreSQL), and DynamoDB for building highly available data pipelines supporting real-time banking data feeds.
- Designed MongoDB-based document storage for storing digital loan documents, e-statements, and KYC-related metadata.
- Developed role-based access control (RBAC) mechanisms within Flask APIs to segregate permissions for internal users, auditors, and branch managers.
- Automated infrastructure provisioning for banking applications using Terraform and CloudFormation, ensuring auditability and repeatability.
- Designed internal admin dashboards with Flask, Bootstrap, and REST APIs to support risk analysts, support teams, and financial advisors.
- Deployed monitoring and alerting for mission-critical banking services using ELK stack, AWS CloudWatch, and Datadog.
- Created Jenkins CI/CD pipelines for zero-downtime deployment of online banking platforms, ensuring rollback mechanisms and blue-green deployments.
- Wrote unit and integration test cases using Pytest and Selenium for high-risk modules such as funds transfers, customer authentication, and loan approvals.
- Implemented data encryption at rest and in transit for all sensitive banking data using AWS KMS and secure coding practices.

- Supported internal audit requirements by generating compliance reports from MongoDB and PostgreSQL using SQL Alchemy and Python.
- Developed interactive dashboards and analytics tools for bank managers using Flask APIs, D3.js, and Tableau integration.
- Designed secure and reproducible container environments with best practices in dependency and secrets management.
- Enhanced mobile and web app security with multi-factor authentication (MFA) and biometric integration using third-party identity providers.
- Integrated real-time alerts and customer notifications using AWS SNS, SES, and in-app push notification APIs.
- Performed data reconciliation scripts between customer accounts, transaction logs, and external financial systems using Python and SQL.
- Provided technical documentation, training, and handover to internal support teams on new banking APIs and tools.
- Optimized banking APIs to reduce latency and improve throughput using Redis caching and connection pooling strategies.

Environment: Python 3, Flask, Pyspark, DynamoDB, AWS (S3, Lambda, RDS, KMS, SNS, SES, CloudWatch), Docker, Kubernetes (EKS), Jenkins, Terraform, CloudFormation, Git, SQL Alchemy, Pytest, Redis, ELK Stack, OAuth 2.0, JWT, D3.js, Tableau, RESTful APIs.

Client: CareSource, Dayton, OH

Jun 2021- Sep 2022

Sr. Python Developer

Responsibilities:

- Managed datasets using Panda data frames and MySQL, queried MYSQL database queries from python using Python-MySQL.
- Developed **RESTful microservices** in Python/Flask and designed front-end components with **ReactJS** and Bootstrap.
- Designed and developed ETL integration patterns using PySpark, AWS glue.
- Worked with developing backend services with NodeJS and Python in AWS Lambda for creating microservices.
- Developing server-based web application to determine patient journey using Flask.
- Worked extensively on AWS Components such as Elastic Map Reduce (EMR), Elastic Compute Cloud (EC2), and Simple Storage Service (S3).
- Created Athena and integrated with AWS Glue to fully manage ETL service that can categorize the data.
- Experience in designing and deploying AWS Solutions using EC2, S3, EBS, Elastic Load balancer (ELB) and auto scaling groups.
- Used Bootstrap front-end framework for faster and easier web development with MySQL.
- Extensively worked on Jenkins to implement continuous integration (CI) and Continuous deployment (CD) processes.
- Build and test functionality within a production pipeline.
- Involved in data ingestion into MySQL using Kafka - MySQL pipeline for full load and Incremental load on variety of sources like web server, RDBMS, and Data APIs.
- Involved in ingestion, transformation, manipulation, and computation of data using MySQL, and Spark.
- Responsible for creating tables and MySQL pipelines which are automated to load the data into tables from Kafka topics.
- Used Boto3 to create and update IAM users, policies, and roles, automating user onboarding and access control.
- Created AWS Lambda architecture to monitor AWS S3 Buckets and triggers for processing source data.
- Analyzed the SQL scripts and designed the solution to implement using Pyspark
- Built micro services using API gateway.
- Built Lambda functions using Boto3 to read/write to DynamoDB, enabling scalable, serverless data handling.
- Worked on ETL services by developing and deploying AWS Lambda functions for generating a serverless data pipeline which can be written to Glue Catalog and can be queried from Athena.
- Experience in Real time streaming the data using Spark with Kafka.
- Experience working on Vagrant boxes to setup a local Kafka and Stream Sets pipelines.
- Maintained version control of Python by using GIT. (Merge Branch, Push, and Commit).

Environment: Python 3, Django, AWS, Kafka, AngularJS, CSS, PostgreSQL, MySQL, JSON, HTML, NoSQL, JavaScript, AWS, Airflow, UNIX and Windows.

Client: USAA, Plano, TX

Feb 2019 – May 2021

Role: Python Developer

Responsibilities:

- Handled operations and maintenance support for AWS cloud resources which includes launching, maintaining, and troubleshooting EC2 instances, and S3 buckets, Virtual Private Clouds (VPC), Elastic Load Balancers (ELB) and Relational Database Services (RDS). Used Amazon EMR for map reduction jobs and test locally using Jenkins.
- Developed customer-based features and full-stack applications using **Python (Django)** and **ReactJS/Angular**, ensuring responsive user interfaces and secure API integration.
- Utilized PyQt to provide GUI for the user to create, modify and view reports based on client data.
- Used PyQuery for selecting DOM elements when parsing HTML.
- Developed functions and features and integrated the APIs in Angular 5/7. Used Angular Services for consuming REST API using HTTP/AJAX libraries.
- Used Angular Routing Module to implement routing, and role-based routing for Single Page Application.
- Used Django configuration to manage URLs and application parameters. Involved in development of Web Services using REST for sending and getting data from the external interface in the XML format.
- Wrote Python scripts to manage AWS resources from API calls using Boto3 SDK and worked with AWS CLI.
- Built numerous Lambda functions using Python and automated the process using the event created.
- Created an AWS Lambda architecture to monitor AWS S3 Buckets and triggers for processing source data.
- Wrote Python scripts to manage AWS resources from API calls using BOTO3 SDK.
- Worked on ETL Migration services by developing and deploying AWS Lambda functions for generating a serverless data pipeline which can be written to Glue Catalog and can be queried from Athena.
- Used AWS data pipeline for Data Extraction, Transformation and Loading from homogeneous or heterogeneous data sources and built various graphs for business decision-making using Python matplotlib library.
- Built ETL pipeline for data ingestion, data transformation and data validation on cloud service AWS, working along with data steward under data compliance.
- Developed Restful Microservices using Django and deployed on AWS servers using EBS and EC2.
- Designed high availability environment for Application servers and database servers on EC2 by using ELB and Autoscaling. Used AWS Lambda to run code virtually.
- Used IAM for creating roles, users, and groups and also implemented Multi-Factor Authentication (MFA) to provide additional security to AWS accounts and its resources.
- Added support for Amazon AWS S3 and RDS to host static/media files and the database into Amazon Cloud.
- Designed and developed ETL Processes in AWS Glue to migrate campaign data from external sources like S3, parquet/Text Files into AWS Redshift.
- Wrote functional API test cases for testing REST APIs with Postman and integrated with Jenkins server to build scripts.
- Developed Microservices by creating REST APIs and used them to access data from different suppliers and to gather network traffic data from servers.
- Implemented code coverage and unit test plug-ins with Maven in Jenkins, using GIT version control system.
- Pulled information from Jira using REST API and Python to populate excel files for management reports.
- Responsible for developing and creating unit test cases using Python unit testing frameworks (PyTest).
- Worked on Jira for managing the tasks and improving the individual performance.

Environment: Python 3.5, Django 2.2, AWS Cloud Services – EC2, S3, VPC, ELB, Lambda, Autoscaling, IAM, EMR, CloudFront, HTML, CSS3, Angular 5/7, PyQuery, PyCharm, REST API, Maven, Jenkins, GIT, PyUnit, Tomcat, Windows, JIRA.

Client: HSBC, India

Aug 2014 – Nov 2017

Python Developer

Responsibilities:

- Wrote Python scripts to parse XML documents and load the data in database.
- Developed entire frontend and backend modules using Python on Django Web Framework by implementing MVC architecture.
- Used Python Django to interface with jQuery UI and manage storage and deletion of content.

- Used Python's modules NumPy, Matplotlib, etc. for generating complex graphical data, creation of histograms etc.
- Developed views and templates for applications with MySQL, Python and Django's view controller and templating language to create a user-friendly interface to perform in a high-level.
- Used Python based GUI components for the front-end functionality such as selection criteria.
- Received business logic specifications and design documents, and painted HTML using CSS and business logic in Angular 4/2 and JavaScript.
- Managed large-scale, geographically distributed database systems, including relational (Oracle, MySQL Server, PostgreSQL) and NoSQL (MongoDB) systems.
- Prepared standard reports, charts, graphs, and tables from a structured data source by querying data repositories using Python and SQL. Upgraded the existing UI with HTML, CSS, jQuery and Bootstrap.
- Used MySQL as backend database and MySQL dB of python as database connector to interact with MS SQL Server.
- Created AJAX calls using AngularJS to handle data operations including ordering and filtering.
- Used PyCharm as Python IDE. Used Maven as the build tool and refactored POM file.
- Managed code versioning with SVN and deployment to staging and production servers.
- Designed database schemas and data models optimized for Azure SQL, considering performance and scalability requirements.
- Implemented database constraints, indexes, and relationships through Python scripts and SQL scripts.
- Used Jenkins plugins for code coverage & to run the entire test before generating WAR files.
- Developed Oracle objects such as Tables, Views, Indexes, Procedures, Functions and Packages in PL/SQL.
- Performed Unit testing and developed the unit test cases using PyUnit framework.
- Used Jira (software development tool) for planning, tracking and reporting daily as well as monthly goals.

Environment: Python 3.2, Django, NumPy, Matplotlib, Azure, REST API, MySQL, Oracle, PostgreSQL, MongoDB, HTML5, CSS3, Angular 4/2, JQuery, Bootstrap, WebLogic, PyCharm, SVN, Jenkins, PyUnit